

Extra data sources — what we checked

We reviewed **Bloomberg** for news, and **Polymarket**, **Kalshi** and **ForecastEx** for event probabilities. The expanded Polymarket review now separates the US and international services. **Polymarket US works technically for a read-only integration; a local MVP is implemented and tested. Company usage rights and employee production access controls remain to be resolved.** ForecastEx remains another candidate to evaluate. Other provider sections retain the earlier review.

Bloomberg — no

Bloomberg has the best data. **The blocker is their contract, not the price.**

Their licence, which we read directly, forbids storing their data in a database, sending it to anyone else, using it in an automated system, or connecting from a machine not logged into a Terminal. Our service does all four. A seat costs **\$31,980/year**, and buying one wouldn't help — the restrictions come with it.

They sell feeds needing no Terminal, but those are for desks to trade on, not to show readers. The one product that *allows*

republishing is a separate media business selling by topic ("Technology", "Energy"), not by company — so even with money, it can't answer "news about this company."

This won't change: their licence bars anything that could cost them a Terminal subscription.

Polymarket — US integration is feasible; company usage rights remain unresolved

Review updated: 9 September 2026; saved API evidence is from 8 September. The proposed production use is an employee-only analytics panel beside our news feed, with no trading. **Polymarket US and international Polymarket are separate products:** their APIs, accounts, markets and legal routes must be assessed separately. A US account does not provide the international Gamma/CLOB catalogue. Public US endpoints and the local preview have been tested. Private account credentials and authenticated upstream streaming have not been inspected or tested.

1. What we verified on Polymarket US

Direct requests from this machine succeeded **without a VPN, login or API key**. This corrects the earlier network conclusion for the US service; deployment-server connectivity still needs its own check.

Read-only check	Observed result on 8 September
GET /v1/markets? limit=3&active=true&closed=false	HTTP 200; three active sports contracts with metadata and quotes
GET /v1/markets/{slug}/bbo and /book	HTTP 200; bid/ask prices, quantities and market statistics
GET /v1/search?query=Fed&limit=3	September and October Fed decisions, plus a 2026 rate-hike event, marked active/open
Search for interest rates	Brazil, Mexico and Japan central-bank decision events; a Mexican contract also returned a nonempty book
GET /v1/price-history for one sports contract	HTTP 200; the one-hour window was empty and the one-day window contained one sample

The base is `https://gateway.polymarket.us`. These are small connectivity and schema checks, **not a liquidity survey or a production reliability test**. A general introduction still described finance as forthcoming, but the live API already listed macro markets. Conversely, a search for inflation returned no results; that alone does not prove the entire catalogue lacks inflation markets. Public API, market listings, search.

A later saved Fed sample included five contracts, event artwork and 249 daily history points / 57 weekly points, with visible gaps. The dedicated inventory records exact request times, response fields and screenshots; those observations must not be substituted for current quotes. Data inventory and local MVP record.

2. Connection options and freshness

Route	Data and access	Fit for our platform
Public REST: <code>gateway.polymarket.us</code>	Markets/events, search, images, quotes, books, settlement and history; no key	Local preview route; confirm company scope before recurring production use
Retail market WebSocket: <code>wss://api.polymarket.us/v1/ws/markets</code>	API-key authentication; real-time books, prices and trades	Optional connection implemented; credentials and upstream streaming not tested
Direct Exchange data access	Read-only REST/gRPC for reference data, BBO, depth and statistics; provisioned credentials	Explicitly documented route for research teams and analytics platforms

The public API documents **20 requests/second/IP**. REST gives a snapshot per request; the local MVP's **10-second polling** does not capture every intervening change. History requests have their own cache and sampling intervals. The retail WebSocket supports up to 100 markets per subscription and optional batching. Neither a successful request nor the phrase real time establishes a latency guarantee. We have not measured the authenticated stream. Public limits, WebSocket, data guide.

Retail keys require app signup and identity verification before developer-portal key generation; the retail route is described as personal accounts only. This requirement does not apply to anonymous public REST. Institutional data-only streaming uses separate credentials/scopes and says trading KYC and a participant ID are unnecessary; contractual onboarding still applies. Personal login is therefore neither necessary for public reads nor sufficient evidence of company entitlement. Retail authentication, API types, institutional streaming.

Institutional limits differ materially: book/BBO reads are documented at 12/minute/firm, with streaming available for frequent updates. **Do not reuse the public REST polling budget for an institutional connection.** Exchange limits.

3. What is allowed, and what remains unconfirmed

The public documentation expressly invites market-display applications. However, the linked US app terms include APIs and grant personal, noncommercial data use connected with trading (§5); automation through authorized APIs is allowed (§7). **Our reading: this does not establish a licence for company analytics.** A public endpoint or SDK software licence does not settle that question. US terms, effective 25 September 2025.

The following is a scope checklist for our proposed use, not a claim that every item requires a separate paid product:

Proposed activity	Position for this project
Access documented public endpoints without a key	Officially supported technically; bounded reads worked
Recurring collection and employee dashboard display	Obtain confirmation that the public route covers our company, or agree another route
Store snapshots, raw responses, backups and charts	Specify retention, historical use and deletion obligations; no project entitlement established
Display event images or copy article content	Confirm applicable content/image rights; a URL or public article is not an unrestricted reuse licence
Compute changes, sentiment or cross-source scores	Confirm derived-data and combination rights
Send data to Anthropic or another processor	Separate unresolved use; employee-only display does not establish this permission
Public pages, customer API, downloadable exports	Outside the current scope; require separate rights assessment
Scrape website pages	Unnecessary; scraping and bulk downloads are restricted by §5

Polymarket US has a documented **Market Data Agreement** route via **data@polymarket.us**, followed by public-key submission, review and issued credentials. Its actual agreement was not available in the public material reviewed, so retention, permitted recipients, AI processing, price and termination terms remain unknown. This is one route to resolve rights; **we have not established that every public API request requires it, or that payment is always required.** Data onboarding.

Public REST connects without contacting Polymarket; mandatory payment has not been established for our use. Neither an interactive local preview nor an API key establishes company entitlement. No blanket local-research or pilot exemption was found. The July 2026 Institutional Research announcement offers commercial data sales for both US and international markets; it supplies no public price for our project. Published US trading fees are not data-licence prices. Institutional announcement, Trading fees.

The institutional guides recommend local history storage, reference-data caching and custom candle aggregation. These are intended capabilities of that route, not activities that are categorically unavailable. The agreement must still define our retention and commercial use; no specific permission for LLM prompts, embeddings or model training was found in the reviewed sources. Storage practices, Custom aggregation.

A trading-platform/ISV partnership is a different route with its own agreement; it should not be assumed to authorize our news-and-data analytics use. Likewise, participant agreements concern exchange participation and should not be applied automatically to an anonymous public-data reader. ISV route, participant agreement.

4. Local MVP and the production architecture

Implemented local preview: Next.js /polymarket page → server-side adapter → public Polymarket US REST. It combines search/topics, event images, YES/NO prices, a time-based history chart with shaded gaps, book context, statistics, rules and separately sourced article cards. Selected quotes/book refresh every **10 seconds**; history is cached for 30 seconds, event data refreshes every 60 seconds and news every five minutes. Hidden tabs and the pause control suspend polling. The product AI panel is hidden and prices do not enter its AI pipeline. Optional server-side WebSocket support forwards status, quote and book updates through SSE. No keys are configured locally and no authenticated stream has been verified. Eleven browser checks completed on **9 September at 02:27:04 UTC**, including actual data, refresh, pause/resume, selection, chart interaction, labelled failure handling, mobile and dark layouts. Nineteen unit tests, TypeScript, targeted ESLint and the production build passed. Real local screenshots, the separate **243-point/four-gap** chart capture and exact times are in the inventory.

News comes from a separate source. Polymarket's API descriptions are contract rules, not article bodies. Polymarket does publish journalism in **The Oracle**, and its international Journalism Tools offer news alongside market timelines; no supported news/article endpoint was found in the US API index. The MVP uses our existing news sources for **Related news**, preserving publisher attribution and links. Topic relevance does not demonstrate that an article caused a price move. Event image URLs are available from US responses; their content rights must be included in the production scope. The Oracle, Journalism Tools, US API index.

The tested news request returned nine results with eight image URLs and six visible cards. Its newest article was dated **30 August 2026**; working provider access does not establish real-time news delivery.

Future production flow: Polymarket US API → one backend collector → dedicated prediction tables in Postgres → authenticated internal API → employee panel beside macro news. The local preview does not implement this persistent, employee production deployment.

Our standalone news-service already uses TypeScript, Fastify and Postgres, with ingestion jobs and an optional scheduler disabled by default. Its provider interface returns articles linked to company listings; prediction contracts need a separate module rather than an entry in NEWS_PROVIDER_ORDER. The Next.js application already consumes this service over HTTP through src/lib/newsService.ts.

Production component	Proposed change after the preview
Collector	Add a dedicated job and lock; begin with 20 selected macro contracts, metadata every 15 minutes and BBO every 30–60 seconds
Storage	Add market definitions, outcome identities, snapshots and ingestion status; retain source product, upstream IDs, slugs, rules and both upstream/collection timestamps
Internal API	Add market-list/detail/history/status endpoints; serve stored state so each page view does not independently call Polymarket
Interface	Add an optional Event markets panel to GlobalNewsPage, with source link, exact question/outcome, quote basis, age, status and spread
Operations	Track errors, 429s, missing/stale quotes and last successful collection; a failed prediction panel must not break news

For that separate production design, twenty contract requests every 30 seconds average **0.67 requests/second** before metadata, history and book calls. This is a proposed capacity estimate, not a licence or freshness promise. Stagger requests and share the budget across all workers using the same public IP. Start with selected books on demand; streaming is not needed simply to put market context next to news.

Requirements before employee production deployment:

- **Enforce employee access.** The root app describes itself as single-user with no authentication. The backend token currently protects fetch operations, not data reads. Put both the panel and its API behind employee authentication and a restricted service boundary. CORS or hiding a page does not enforce this scope.
- **Use an actual per-second limiter.** The existing per-minute token bucket starts full; setting it to 1,200/minute would permit a burst incompatible with 20/second.
- **Preserve decimals.** The backend globally converts Postgres `NUMERIC` values to JavaScript numbers. Use validated decimal strings or integer ticks for prices; confirm quantity and notional units before calculation.
- **Keep signing keys on the server.** The existing browser-side live-stock-price pattern must not be copied for a Polymarket signing secret. Any stream collector belongs in a supervised backend process with reconnect and snapshot recovery.
- **Keep the initial feature out of AI prompts.** The existing research pipeline sends inputs to Anthropic. Enable this separately only after its data-processing and derived-output scope is confirmed.

Code inspected: backend `src/providers/types.ts`, `src/ingest/scheduler.ts`, `src/util/rateLimiter.ts`, `src/db/pool.ts`, `src/api/server.ts` and `src/api/routes/operations.ts`; frontend `README.md`, `src/lib/newsService.ts`, `src/components/GlobalNewsPage.tsx`, `src/lib/useLiveQuote.ts`, `src/lib/pipeline.ts` and `src/lib/claude.ts`.

5. Data interpretation and checks before rollout

A contract price is a market-implied estimate, not a verified event probability. Keep bid, ask, last trade and display price separate. Show spread, missing sides, last-trade age and contract status; do not silently substitute zero for missing data. A Fed contract is macro context, not automatically company-specific news. Only combine outcomes after checking that their resolution rules make them mutually exclusive and exhaustive.

The price-history endpoint provides **book-derived YES/NO display prices**, not executed-trade history. YES normally follows the best ask and NO the complement of the best bid, so their sum can exceed one. Fixed windows use different sampling intervals; cache identical history requests for at least 30 seconds. Label provider history separately from any snapshots collected by our own job. The initial sports sample was empty/sparse; a later Fed sample had 249 daily and 57 weekly points with gaps. Neither establishes a complete backfill for every contract. History reference.

One actual listing returned deprecated `outcomes / outcomePrices` arrays whose apparent order disagreed with explicit `marketSides`. Use validated side IDs, `long`, descriptions and quote fields; **do not blindly zip the legacy arrays**. The official TypeScript SDK supports public reads, but the reviewed market types omit `marketSides` and its market resource has no history helper. Prefer a small validated REST adapter initially; if using the SDK, pin a version and validate responses rather than relying on its types alone. Market schema, SDK types, SDK market resource.

Keep market closure and final settlement separate. A missing settlement response can mean the market is not settled; some contracts permit alternative settlement values. Preserve the exact rules and state instead of assuming every closed market has resolved to zero or one. Settlement endpoint, Settlement rules.

A sampled Mexican rate-decision contract had a bid of 0.98 and ask of 0.99, with only 19.17 units displayed at the best bid; its last trade was from the previous day. This is one observation, not an overall liquidity finding. An approved pilot should measure spread, depth, trade age, missing observations and history coverage for each selected contract before treating the feed as a useful signal.

Verification should cover outcome mapping, decimal units, sparse history, closed/resolved markets, retries and 429s, duplicate collection, stale labels, unauthenticated access rejection and keeping data out of AI/export paths. If the optional stream is enabled, validate actual message shapes and recovery after a gap; overview and detailed examples currently differ. Do not promise lossless replay without testing the protocol. WebSocket overview, detailed market stream.

6. What we can do now and what needs a decision

1. **Use the validated local preview:** selection, prices, images, charts, 10-second REST refresh and separately sourced news have been checked; actual captures and failure handling are recorded. Synthetic fixtures remain useful for edge cases; a working preview does not establish production rights. Keep unattended production collection disabled.
2. **Resolve the exact use:** ask whether our company may use the public gateway for recurring internal analytics, or needs a Market Data Agreement. Specify legal entity and country, employee/hosting locations, audience, fields, retention, backups, images, derived metrics and any external processors. Request permitted route, fees/trial terms and deletion obligations. No request has been sent and no agreement or paid access has been accepted.
3. **Run an approved REST pilot:** proposed duration one week, 20 relevant contracts, private employee view and measured quality. Test connectivity from the intended deployment environment. The period and watchlist are design choices, not provider entitlements.

4. **Decide on production:** proceed only with documented usage scope and useful coverage/freshness. Add streaming if measurements justify it; consider AI use as a separate scope extension. Licence price is unknown, and public access alone is not evidence of a zero-cost commercial entitlement.

7. International Polymarket remains a separate option

For `polymarket.com`, the checked direct connections reset; a remote reader returned a Gamma market and CLOB clock. This establishes limited retrieval, not a dependable deployment path. Its Gamma, CLOB and WebSocket interfaces must not be mixed with the US APIs or account credentials. International data APIs.

We retrieved the actual embedded terms automatically. The 11 August 2026 version, §4.2, restricts access/use by covered capital-markets entities, including specified fintech businesses and data distributors, and separately restricts redistribution. It covers derived/aggregated data and contains no explicit internal-pilot exception. Our legal entity's activities determine applicability. §4.1 restricts VPN circumvention; changing network routes does not resolve rights. International terms.

ICE announced exclusive distribution for institutional capital markets, and international Polymarket directs relevant firms to its licensing team and ICE. Contact: data-licensing@polymarket.com. Confirm usage scope, delivery and pricing; this is not necessarily buying an API key. **Do not transfer the ICE conclusion to Polymarket US:** the US documentation provides its own direct data-onboarding route. Neither route's price has been established for our project. ICE announcement, international licensing.

Kalshi — no

A properly regulated US exchange, so it looked promising. But its developer agreement, which we read directly, limits the API to *"facilitating a member's own trading"* and forbids *"collecting, caching, aggregating, or storing data."*

Plainly: **their API is for you to trade with, not to build a product on.** Showing their prices to users is exactly what it prohibits. Also network-blocked.

ForecastEx — worth trying

This one works. ForecastEx is a US exchange regulated by the CFTC, and that regulation **requires it to publish its daily prices.** It does, as plain CSV files on its website.

We downloaded them: no key, no account, current data. Each file gives, per market per day, open, high, low, close, settlement price and open interest — a full history of how the odds moved. It covers inflation (CPI), Fed rate decisions, payroll employment, jobless claims, GDP, retail sales, housing starts and elections.

Rights still need checking: the earlier review identified these files as regulatory disclosures and did not locate applicable reuse terms. Required publication does not itself establish permission for our integration.

Check first: most actual trading there is in *weather* markets. The economic markets are real, but we haven't measured whether enough money trades in them for the numbers to mean much.

Suggestion

Evaluate **Polymarket US alongside ForecastEx** for an employee market-context panel. The verified US API and live macro catalogue change the earlier basis for prioritizing ForecastEx alone. Keep the news pipeline intact and use a separate prediction-data module.

Use the validated local US preview while confirming its exact corporate usage scope. Then run a small approved employee REST pilot and measure data quality. ForecastEx still needs usage-rights and liquidity checks; publication of data alone is not a blanket reuse licence. International Polymarket requires its own access/licensing assessment. The local preview is not a production deployment; no licensing outreach was performed.